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Math 208 Final Exam Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The rank of the matrix $A = \begin{bmatrix} 1 & 3 & -4 & -8 & 6 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 7 & -10 & -19 & 13 \end{bmatrix}$ is

(a) 3

(b) 4

(c) 1

(d) 2

(e) 5

2. The solution of the linear differential equation $x\dfrac{dy}{dx} - 3y = x^3$ is

(a) $y(x) = x^4 \ln x + cx^2$

(b) $y(x) = x^2 \ln x + cx^2$

(c) $y(x) = x^3 \ln x + cx^3$

(d) $y(x) = x^3 \ln x + cx$

(e) $y(x) = x^4 \ln x + cx^4$

3. A particular solution of the differential equation $y'' + 4y' - 5y = 8e^{3x}$ is

(a) $y_p = e^{3x}$

(b) $y_p = \dfrac{1}{2}e^{3x}$

(c) $y_p = \dfrac{1}{4}e^{3x}$

(d) $y_p = \dfrac{1}{8}e^{3x}$

(e) $y_p = \dfrac{1}{3}e^{3x}$

4. The general solution of the differential equation

$$(y - 2x)\frac{dy}{dx} = 6$$

is

(a) $y + 3\ln|y - 2x - 3| = c$

(b) $y - \ln|y - 2x + 4| = c$

(c) $x + \ln|y - 2x + 3| = c$

(d) $x - \ln|y - 2x + 4| = c$

(e) $(y - 2x) + \ln|y - 2x + 3| = c$

5. A possible fundamental matrix for the system $X' = \begin{bmatrix} 6 & 2 \\ 5 & 3 \end{bmatrix} X$ is

(a) $\Phi(t) = \begin{bmatrix} -2e^t & e^{8t} \\ 5e^t & e^{8t} \end{bmatrix}$

(b) $\Phi(t) = \begin{bmatrix} e^t & -2e^{8t} \\ e^t & 5e^{8t} \end{bmatrix}$

(c) $\Phi(t) = \begin{bmatrix} 2e^t & e^{6t} \\ -5e^t & e^{6t} \end{bmatrix}$

(d) $\Phi(t) = \begin{bmatrix} 3e^t & e^{8t} \\ e^t & 2e^{8t} \end{bmatrix}$

(e) $\Phi(t) = \begin{bmatrix} e^t & e^{8t} \\ e^t & e^{8t} \end{bmatrix}$

6. The general solution of the differential equation

$$D^2(D^2 - 6D + 13)\, y = 0$$

is given by

(a) $y = c_1 + c_2x + c_3e^{2x}\cos(3x) + c_4e^{2x}\sin(3x)$

(b) $y = c_1 + c_2x + c_3e^{3x}\cos(2x) + c_4e^{3x}\sin(2x)$

(c) $y = c_1 + c_2x^2 + c_3e^{3x}\cos(2x) + c_4e^{3x}\sin(2x)$

(d) $y = c_1x + c_3e^{2x}\cos(3x) + c_4e^{2x}\sin(3x)$

(e) $y = c_1 + c_2x + c_3\cos(2x) + c_4\sin(2x)$

7. The solution of the exact differential equation

$$(5x^4 - 4x^3y^3)\,dx + (5 - 3x^4y^2)\,dy = 0$$

is

(a) $x^5 + x^4y^3 + 5y = c$

(b) $x^5 - x^4y^3 + 5y = c$

(c) $x^4 - x^3y^3 = c$

(d) $x^5 - 2x^4y^3 + 3y = c$

(e) $x^4 + x^3y^3 + 5y = c$

8. If $X = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 4 \\ 3 \end{bmatrix} e^{6t}$ is the solution of the initial value problem $X' = \begin{bmatrix} 3 & 3 \\ 4 & 2 \end{bmatrix} X$, $X(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then $c_1 + c_2 =$

(a) 0

(b) $\frac{2}{7}$

(c) $\frac{3}{7}$

(d) $-\frac{1}{7}$

(e) $\frac{1}{7}$

9. An appropriate form of particular solution of the differential equation $y'' + y = x + \cos x$ is

(a) $y_p = Ax + Cx\sin x + Ex\cos x$

(b) $y_p = Ax + B + Cx^2\sin x + Ex^2\cos x$

(c) $y_p = Ax^2 + Bx + Cx\sin x + Ex\cos x$

(d) $y_p = Ax + B + C\sin x + E\cos x$

(e) $y_p = Ax + B + Cx\sin x + Ex\cos x$

10. The minimum radius of convergence of the power series solution for the differential equation $(x^2 - 3)y'' + 2xy' = 0$ about the ordinary point $x = 0$ is

(a) 2

(b) 3

(c) $\sqrt{3}$

(d) ∞

(e) 0

11. Let $u = (1, 2, -3)$, $v = (3, 1, -2)$ and $w = (5, -5, 6)$ be three vectors in $\mathbb{R}^3$. If $w = c_1u + c_2v$, then $c_2 - c_1 =$

(a) -2

(b) 6

(c) 5

(d) 1

(e) 7

12. The largest root of the indicial equation at $x = 0$ for the differential equation $2xy'' - y' - y = 0$ is

(a) $-\frac{1}{2}$

(b) 1

(c) 0

(d) $\frac{1}{2}$

(e) $\frac{3}{2}$

13. A general solution of the system $X' = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} X$ is

(a) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1+t \\ 0 \end{bmatrix} \right) e^{3t}$

(b) $X = \left(c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1+t \\ 0 \end{bmatrix} \right) e^{2t}$

(c) $X = \left(c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1+t \\ t \end{bmatrix} \right) e^{2t}$

(d) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1+t \\ t \end{bmatrix} \right) e^{2t}$

(e) $X = \left(c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1+t \\ t \end{bmatrix} \right) e^{3t}$

14. If $A = \begin{bmatrix} 3 & 0 & -3 \\ 5 & 0 & 7 \\ 3 & 0 & -3 \end{bmatrix}$, then $e^{At} =$

(a) $\begin{bmatrix} 3t & 0 & -3t \\ 5t + 18t^2 & 1 & 7t - 18t^2 \\ 3t & 0 & 1 - 3t \end{bmatrix}$

(b) $\begin{bmatrix} 1 - 3t & 1 & -3t \\ 5t & 1 & 7t \\ 3t & 0 & -3t \end{bmatrix}$

(c) $\begin{bmatrix} 1 + 3t & 0 & -3t \\ 5t + 18t^2 & 1 & 7t - 18t^2 \\ 3t & 0 & 1 - 3t \end{bmatrix}$

(d) $\begin{bmatrix} 1 - 3t & 0 & -3t \\ 18t^2 & 1 & 7t \\ 3t & 0 & -3t \end{bmatrix}$

(e) $\begin{bmatrix} 1 - 3t & 0 & 3t \\ 5t & 1 & 7t - 18t^2 \\ 3t & 0 & 1 - 3t \end{bmatrix}$

15. If the matrix $A = \begin{bmatrix} 9 & -8 \\ 6 & -5 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

(c) $P = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

(e) $P = \begin{bmatrix} 3 & 1 \\ 4 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

16. The matrix $A = \begin{bmatrix} -1 & 0 & 2 \\ 1 & -3 & 0 \\ 0 & 3 & -2 \end{bmatrix}$ has

(a) one eigenvalue of multiplicity 3

(b) no real eigenvalue

(c) one real and one pair of non real eigenvalues

(d) one eigenvalue of multiplicity 1 and one eigenvalue of multiplicity 2

(e) three distinct real eigenvalues

17. The general solution of the system $X' = \begin{bmatrix} -3 & 4 \\ 6 & -5 \end{bmatrix} X$ is given by

(a) $X = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 2 \\ -3 \end{bmatrix} e^{-9t}$

(b) $X = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^t + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} e^{-9t}$

(c) $X = c_1 \begin{bmatrix} 2 \\ -3 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-9t}$

(d) $X = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} e^{-9t}$

(e) $X = c_1 \begin{bmatrix} 2 \\ -3 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{9t}$

18. If the matrix $A = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$ has an eigenvalue $\lambda = 3 + 4i$ with corresponding eigenvector $\begin{bmatrix} 1 \\ -i \end{bmatrix}$, then the general solution of the system $X' = AX$ is

(a) $\begin{bmatrix} c_1 \cos 3t + c_2 \sin 3t \\ c_1 \sin 3t - c_2 \cos 3t \end{bmatrix} e^{4t}$

(b) $\begin{bmatrix} c_1 \cos 4t + c_2 \sin 4t \\ c_1 \sin t + c_2 \cos t \end{bmatrix} e^{3t}$

(c) $\begin{bmatrix} c_1 \sin 4t - c_2 \sin 4t \\ c_1 \cos 4t \end{bmatrix} e^{4t}$

(d) $\begin{bmatrix} c_1 \sin 4t - c_2 \sin 4t \\ c_1 \sin 4t \end{bmatrix} e^{3t}$

(e) $\begin{bmatrix} c_1 \cos 4t + c_2 \sin 4t \\ c_1 \sin 4t - c_2 \cos 4t \end{bmatrix} e^{3t}$

19. The Wronskian of the functions $y_1(x) = x$, $y_2(x) = x \ln x$, and $y_3(x) = x^2$ is

(a) $x \ln x$

(b) $x^2 + x$

(c) 0

(d) x^2

(e) x

20. If $y = \sum_{n=0}^{\infty} c_n x^n$ is a power series solution about the ordinary point $x = 0$ of the differential equation $y'' + xy' + 2y = 0$, then the coefficients c_n satisfy

(a) $c_{n+1} = \dfrac{1}{n+2} c_n,\ n \geq 1$

(b) $c_{n+2} = \dfrac{3}{n+1} c_n,\ n \geq 1$

(c) $c_{n+2} = \dfrac{-1}{n+2} c_n,\ n \geq 1$

(d) $c_{n+2} = \dfrac{-1}{n+1} c_n,\ n \geq 1$

(e) $c_{n+1} = \dfrac{-1}{n+1} c_n,\ n \geq 1$

21. If $y(x)$ is the solution of the initial value problem

$$\frac{dy}{dx} = y^2 \sin x - y^2 \cos x, \qquad y\Big(\frac{\pi}{2}\Big) = \frac{1}{3},$$

then $y(\pi) =$

(a) -2

(b) 0

(c) -1

(d) 2

(e) 1

22. The dimension of the subspace $W = \{(x_1, x_2, x_3, x_4) : x_1 = x_2 + x_3 + x_4\}$ of $\mathbb{R}^4$ is equal to

(a) 1

(b) 4

(c) 2

(d) 3

(e) 5

23. If $y = x^m$ is a solution of the differential equation $x^2 y'' + 5xy' + 4y = 0$, then $m =$

(a) -2

(b) 3

(c) 5

(d) -1

(e) 1

24. In a certain culture of bacteria, the initial amount was 2000. If the number of bacteria doubled after 8 hours, then the number of bacteria present after 24 hours is (Assume the rate of change of population is proportional to the population present at time t)

(a) 8000

(b) 24000

(c) 12000

(d) 16000

(e) 20000

25. The differential equation

$$x^3(1-x)y'' + (3x+2)y' + xy = 0$$

has

(a) two regular singular point

(b) two irregular singular point

(c) no singular point

(d) one irregular singular point and one regular singular point

(e) three regular singular point

26. Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -5 & 8 & -5 \\ 0 & 0 & 3 \end{bmatrix}$. A basis for the eigenspace of A associated with the eigenvalue $\lambda = 3$ of A is

(a) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

(b) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

(c) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

(d) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$

(e) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

27. If $v = \begin{bmatrix} a \\ b \\ 1 \end{bmatrix}$ is an eigenvector of the matrix $A = \begin{bmatrix} 3 & -2 & 0 \\ -1 & 3 & -2 \\ 0 & -1 & 3 \end{bmatrix}$ associated with the eigenvalue $\lambda = 1$, then $a + 2b =$

(a) 4

(b) 2

(c) 6

(d) 0

(e) 8

28. Using variation of parameters, the differential equation $y'' + y = \csc^2 x$ has particular solution $y_p(x) = u_1(x)\cos x + u_2(x)\sin x$. Then $u_2(x) =$

(a) $\sin x$

(b) $-\csc x$

(c) $3\sec x$

(d) $\sec x$

(e) $\tan x$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	Row reduce: new row 2 is $-3\times$ new row 3, so only two nonzero rows: rank 2.
2	**(c)**	$\rho = x^{-3}$: $(x^{-3}y)' = \frac{1}{x}$, so $y = x^3 \ln x + cx^3$.
3	**(b)**	$y_p = Ae^{3x}$: $(9 + 12 - 5)A = 8$, so $A = \frac{1}{2}$.
4	**(a)**	$v = y - 2x$: $vv' = 6 - 2v$, so $v + 3\ln\lvert 6 - 2v\rvert = -2x + C$, i.e. $y + 3\ln\lvert y - 2x - 3\rvert = c$.
5	**(a)**	$\lambda = 1, 8$, eigenvectors $(-2, 5)$, $(1, 1)$: columns $\mathbf{v}_i e^{\lambda_i t}$.
6	**(b)**	Roots $0, 0, 3 \pm 2i$: $y = c_1 + c_2 x + e^{3x}(c_3 \cos 2x + c_4 \sin 2x)$.
7	**(b)**	$F = x^5 - x^4y^3 + g(y)$, $F_y = N$ gives $g = 5y$: $x^5 - x^4y^3 + 5y = c$.
8	**(e)**	$c_1 + 4c_2 = 1$, $-c_1 + 3c_2 = 1$: $c_2 = \frac{2}{7}$, $c_1 = -\frac{1}{7}$, sum $\frac{1}{7}$.
9	**(e)**	$y_c = c_1 \cos x + c_2 \sin x$; $x \to Ax + B$, $\cos x \to x(C \sin x + E \cos x)$.
10	**(c)**	Singular points $x = \pm\sqrt{3}$, so $R \geq \sqrt{3}$.
11	**(e)**	$c_1 = -4$, $c_2 = 3$ (third equation checks), so $c_2 - c_1 = 7$.
12	**(e)**	$r(r-1) - \frac{1}{2}r = 0 \Rightarrow r = 0,\ \frac{3}{2}$; largest is $\frac{3}{2}$.
13	**(c)**	$\lambda = 2$ (double), $v = (1, 1)$, $w = (1, 0)$: $X = (c_1 v + c_2(tv + w))e^{2t}$.
14	**(c)**	$A^3 = 0$: $e^{At} = I + At + \frac{t^2}{2}A^2$ with A^2 having $36, -36$ in row 2.

Q	Ans	Brief explanation
15	**(d)**	$\lambda = 1, 3$ with eigenvectors $(1,1)$, $(4,3)$: $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \mathrm{diag}(1,3)$.
16	**(c)**	$\det(A - \lambda I) = -\lambda(\lambda^2 + 6\lambda + 11)$: roots 0, $-3 \pm \sqrt{2}\, i$.
17	**(a)**	$\lambda = 1, -9$ with eigenvectors $(1,1)$, $(2,-3)$.
18	**(e)**	Re and Im of $e^{3t}(1,-i)^T(\cos 4t + i \sin 4t)$ give the solution.
19	**(e)**	$W = x[2(\ln x + 1) - 2] - [2x \ln x - x] = x$.
20	**(d)**	$(n+2)(n+1)c_{n+2} + (n+2)c_n = 0 \Rightarrow c_{n+2} = \frac{-1}{n+1} c_n$.
21	**(e)**	Separable: $-\frac{1}{y} = -\cos x - \sin x - 2$; at $x = \pi$, $y = 1$.
22	**(d)**	Three free variables x_2, x_3, x_4 give a basis of 3 vectors: $\dim W = 3$.
23	**(a)**	$m(m-1) + 5m + 4 = (m+2)^2 = 0$, so $m = -2$.
24	**(d)**	$P = 2000e^{kt}$, $e^{8k} = 2$, so $P(24) = 2000 \cdot 2^3 = 16000$.
25	**(d)**	Singular points 0 (irregular, xP has $\frac{1}{x^2}$) and 1 (regular).
26	**(c)**	$A - 3I$ gives $x = y - z$: basis $(1,1,0)$, $(-1,0,1)$.
27	**(c)**	$(A - I)v = 0$: row 3 gives $b = 2$, row 1 gives $a = 2$, so $a + 2b = 6$.
28	**(b)**	$W = 1$, $u_2' = \frac{\cos x}{\sin^2 x}$, so $u_2 = -\csc x$.